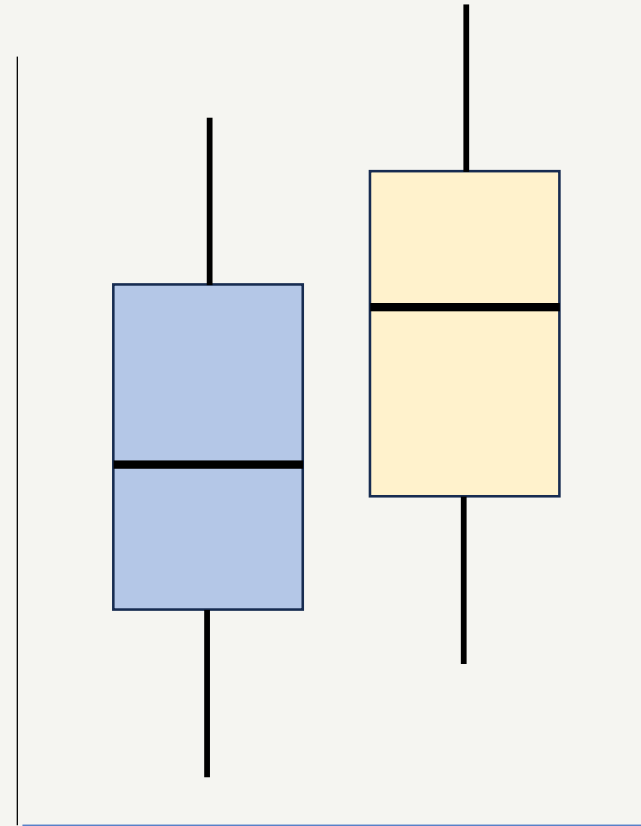


One-Way ANOVA

Dr Metodi Siromahov

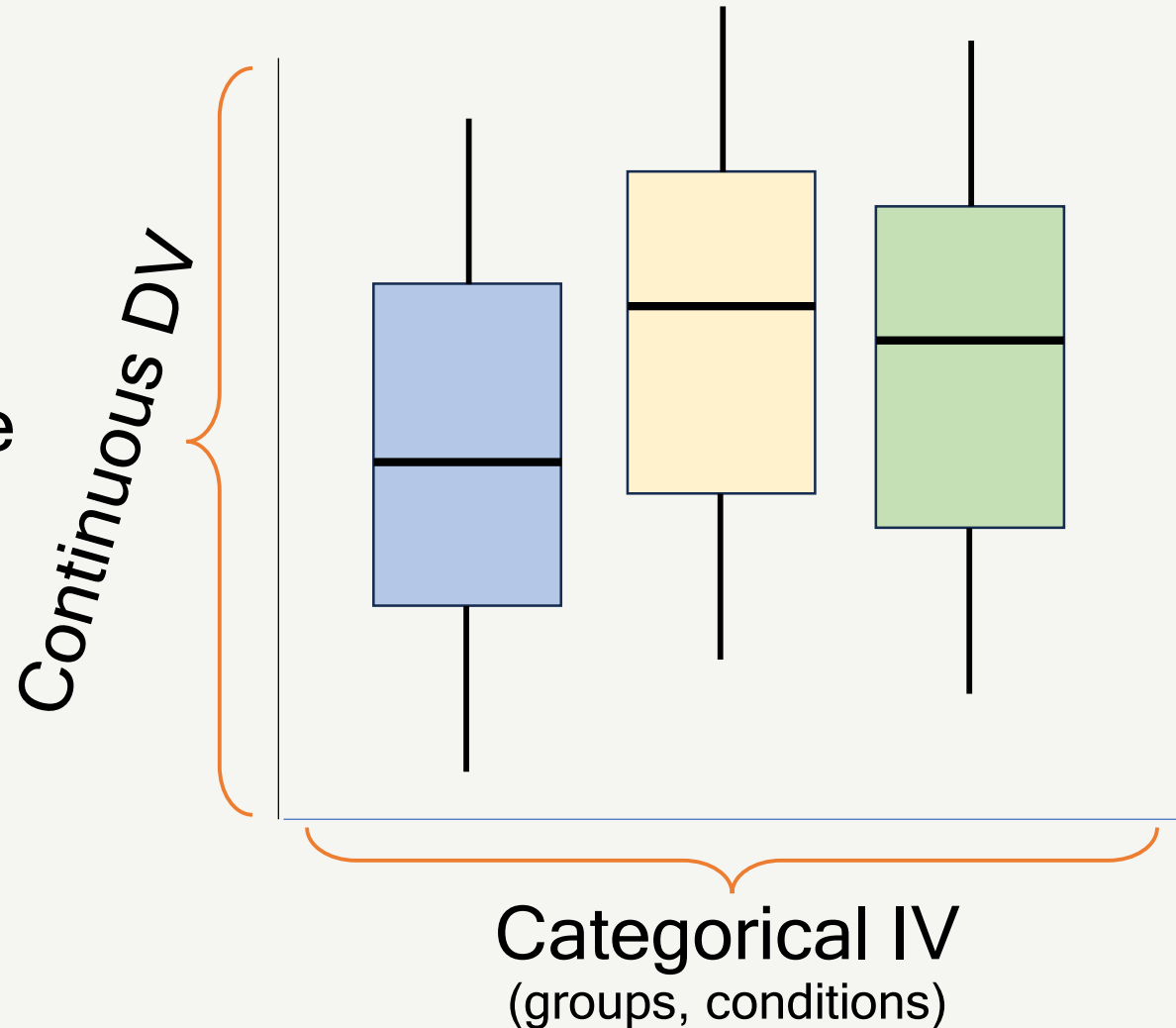
t -tests recap

- t -tests are used when comparing **only two** groups, conditions, or means



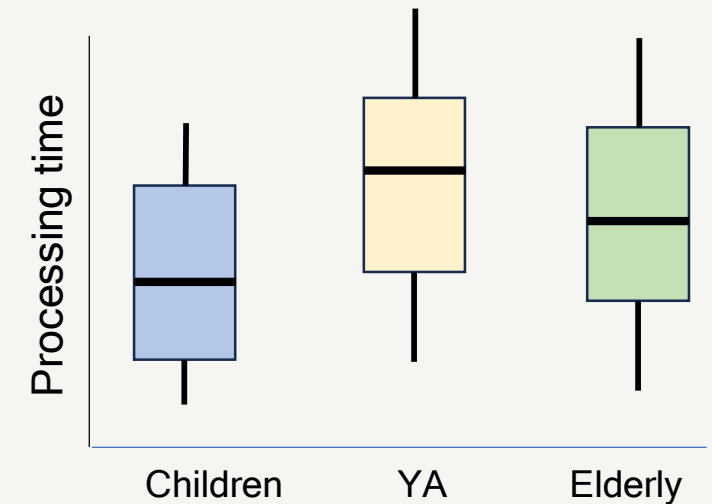
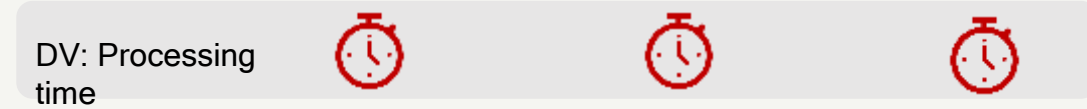
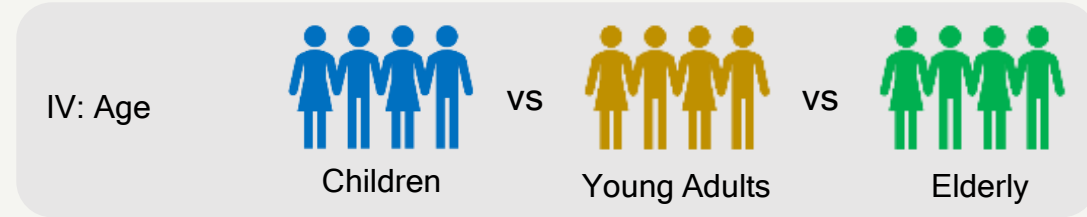
t -tests recap

- t -tests are used when comparing **only two** groups, conditions, or means
- ANOVAs are used when we have more than 2 groups
- **An**alysis **Of** **Va**riance



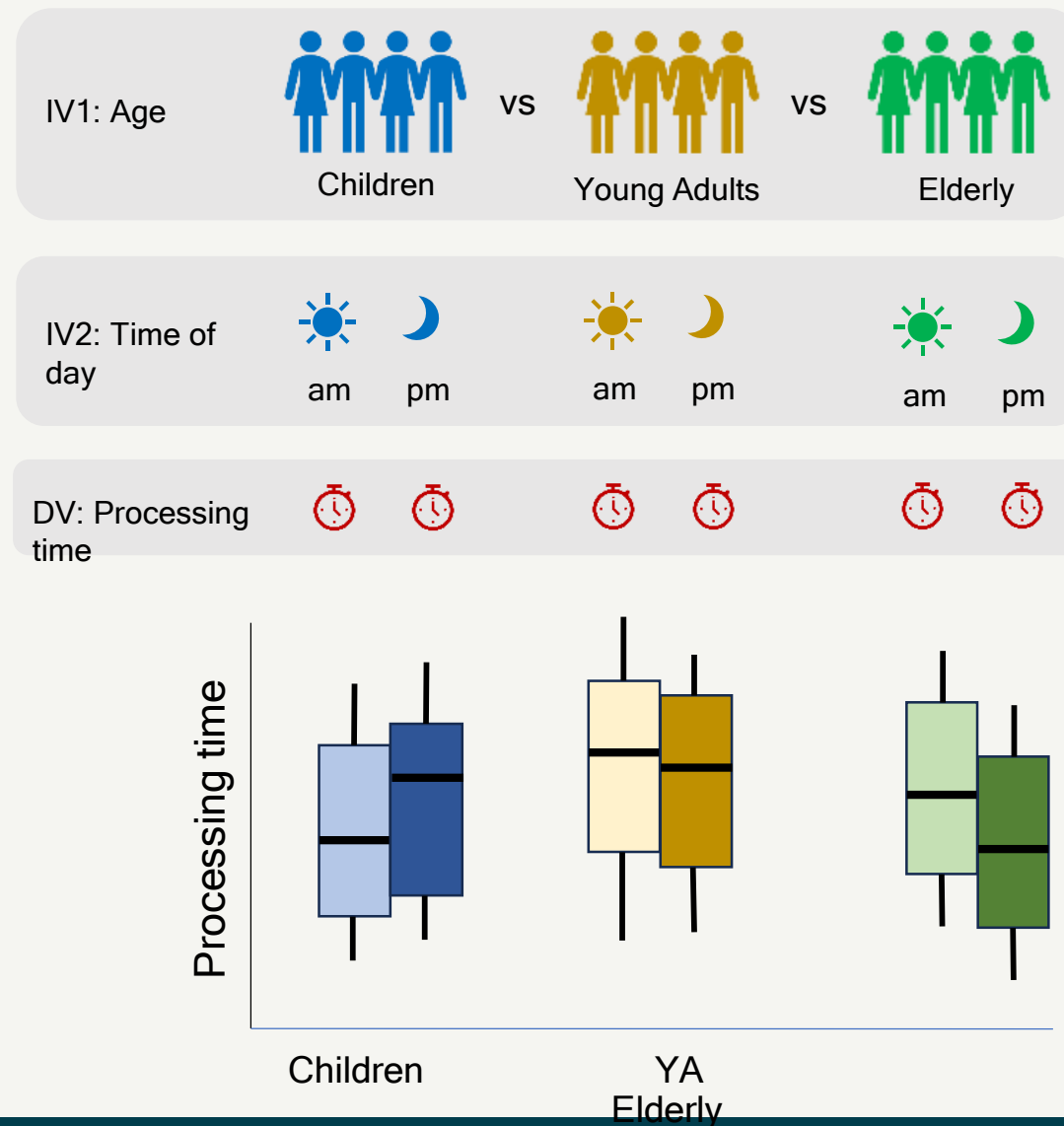
What is an ANOVA?

- Example:
 - IV: age group (children, young adults, elderly)
 - DV: speed of processing



What is an ANOVA?

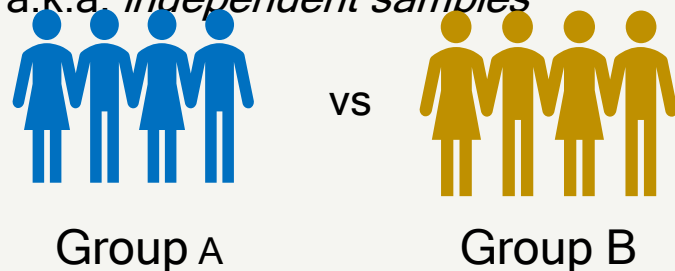
- Example:
 - IV: age group (children, young adults, elderly)
 - DV: speed of processing
- Can have more than 1 IV for multivariate analysis:
 - E.g., does speed of processing depend on age group *and* time of day?
 - IV1: age group (children, young adults, elderly)
 - IV2: time of day (am, pm)
 - DV: speed of processing
- One-way ANOVA = 1 IV
- Two-way ANOVA = 2 IVs → next year



Types of research design

Between-subjects design

Comparing two mutually exclusive groups
a.k.a. *independent samples*



Within-subjects design

The same participants tested at different times
a.k.a. *repeated measures*

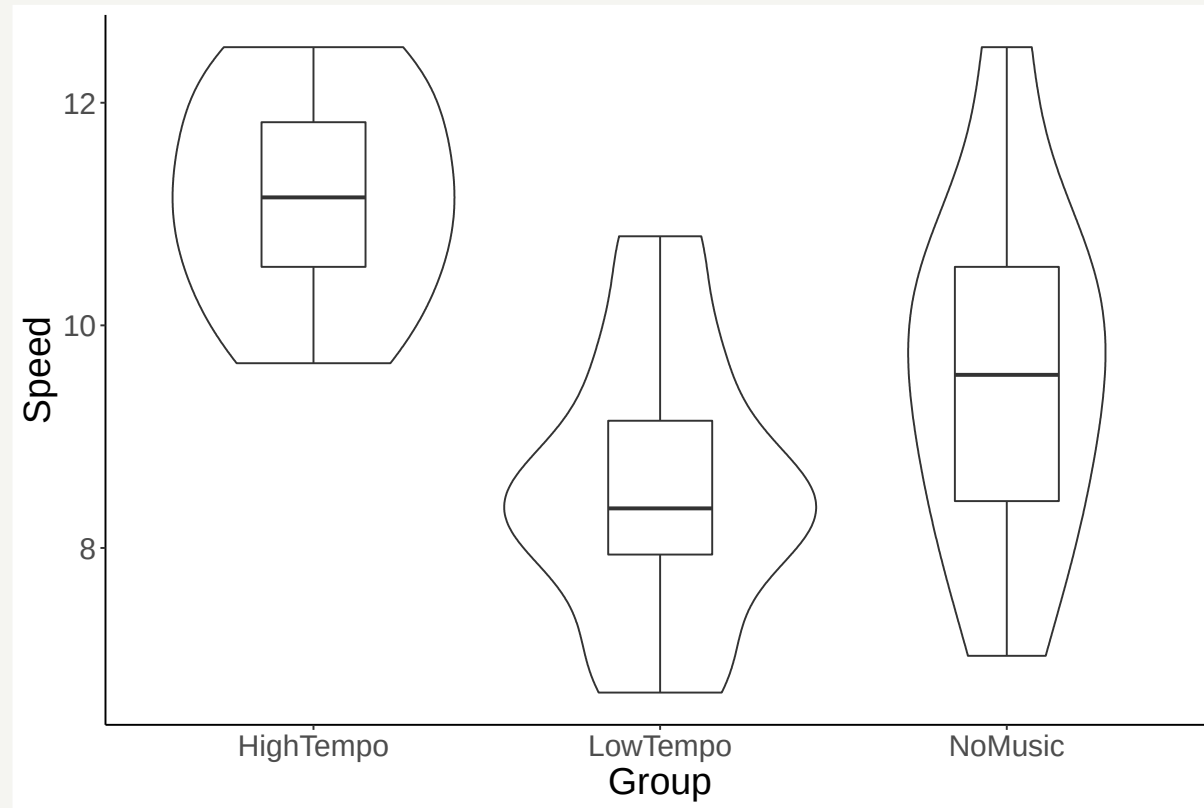


One-way between-subjects ANOVA

- **One-way**: A single categorical predictor with more than 2 levels.
 - Experiment testing the effect of *music tempo* on *10k running speed*
- **Between-subjects** design: 3 different conditions/groups
 1. High Tempo: Participants listened to high tempo, i.e. upbeat, music
 2. Low Tempo: Participants listened to low tempo, i.e. slow, music
 3. No Music: Participants did not listen to any music



Visualise the data



Reminder: Hypothesis testing

Steps:

1. **Determine the null and alternative hypotheses**
2. Choose a test statistic that if extreme would make us doubt null hypothesis
3. Simulate or select theoretical null hypothesis sampling distribution
4. Determine where our observed test statistic lies within this distribution and therefore the probability of it being observed if null hypothesis was true.
5. Check the probability (p-value) against a critical threshold (alpha)

Hypotheses

- **Null hypothesis:** type of music does *not* make a difference to participants' running speed (population means are equal, best described by the grand mean)
- **Alternative hypothesis:** participants will run faster in some conditions than other conditions (the population means are not equal, best described by group-specific means)

Reminder: Hypothesis testing

Steps:

1. Determine the null and alternative hypotheses
- 2. Choose a test statistic that if extreme would make us doubt null hypothesis**
3. Simulate or select theoretical null hypothesis sampling distribution
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5. Check the probability (p-value) against a critical threshold (alpha)

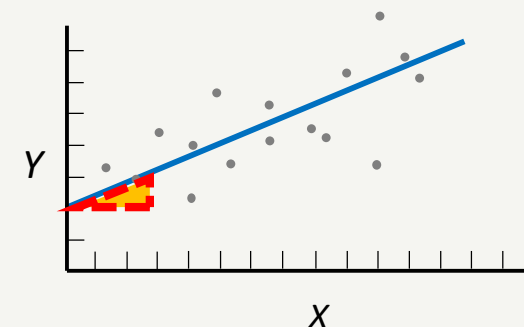
The test statistic: F

- From last week: Building a linear model was only half the task. We also had to test whether the model is a **good fit** for the data.

- We tested goodness-of-fit with an F-test

$$F = \frac{\text{Variance explained by model}}{\text{Remaining (unexplained) variance}} = \frac{MS_{\text{model}}}{MS_{\text{error}}}$$

- We'll do the same thing with the ANOVA



The test statistic: F

$$F = \frac{\text{Variance explained by model}}{\text{Remaining (unexplained) variance}} = \frac{\text{Variance in DV accounted for by group}}{\text{Remaining variance}}$$

- **Large F** = the model explains **a lot** of the variation relative to sampling error.
- **Small F** = the model explains **little** of the variation relative to sampling error.

Remember variance?

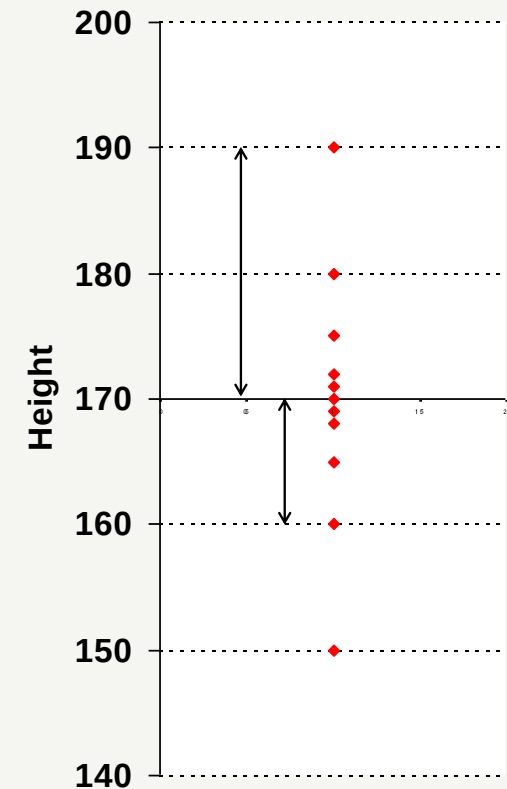
Variance = dispersion of scores around the Mean

= average squared distances of scores from the sample mean

Variance

$$\hat{\sigma}^2 = \frac{\text{sum of squared differences}}{\text{degrees of freedom}} = \frac{SS}{DF} = \frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n - 1}$$

- How much of the variance in the DV is explained by group differences and how much of it is residual?



Does the model fit the data?

$$F = \frac{MS_{model}}{MS_{error}} = \frac{\left(\frac{SS_{model}}{DF_{model}} \right)}{\left(\frac{SS_{error}}{DF_{error}} \right)}$$

Let's find the Sums of Squares first

Step 1: Partition variance

$$SS_{total} = SS_{model} + SS_{error}$$

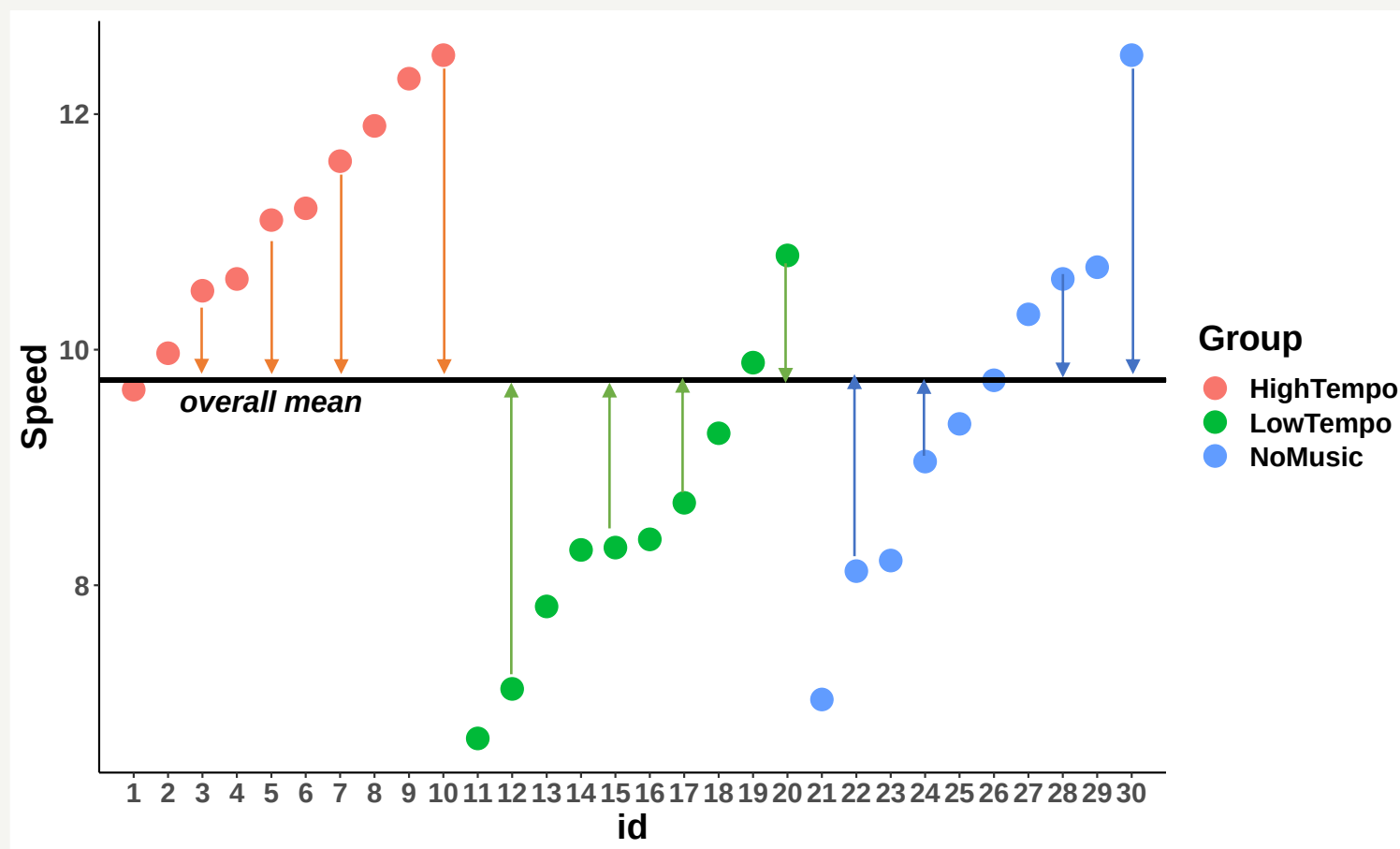
$$data = model + error$$

Sum of squared differences between
each data-point and **overall mean**

Sum of squared of difference between
each data-point and **group mean**

Sum of squared differences between
each group mean and **overall mean**

SS_{total}



Data

	High tempo	Low tempo	No music	
	11.6	8.7	8.21	
	12.3	8.32	9.05	
	9.66	7.82	9.37	
	10.6	8.3	10.3	
	11.2	9.29	10.7	
	10.5	6.7	9.74	
	11.9	9.89	8.12	
	11.1	8.39	12.5	
	9.97	10.8	7.03	
	12.5	7.12	10.6	Overall mean
Group mean	11.13	8.53	9.56	9.74

SS_{total}

Group k	Speed X_{ik}	$[X_{ik} - \bar{X}]$	$[X_{ik} - \bar{X}]^2$
HighTempo	11.6	$11.6 - 9.74 = 1.86$	3.46
HighTempo	12.3	$12.3 - 9.74 = 2.56$	6.55
HighTempo	9.66	$9.66 - 9.74 = -0.08$	0.01
LowTempo	8.7	$8.7 - 9.74 = -1.04$	1.08
LowTempo	6.7	$6.7 - 9.74 = -3.04$	9.24
LowTempo	9.89	$9.89 - 9.74 = 0.15$	0.23
NoMusic	9.37	$9.37 - 9.74 = -0.37$	0.14
NoMusic	10.3	$10.3 - 9.74 = 0.56$	0.31
NoMusic	10.7	$10.7 - 9.74 = 0.96$	0.92
Overall Mean	9.74	$SS_{\text{total}} =$	78.30

Step 1: Partition variance

$$SS_{total} = SS_{model} + SS_{error}$$

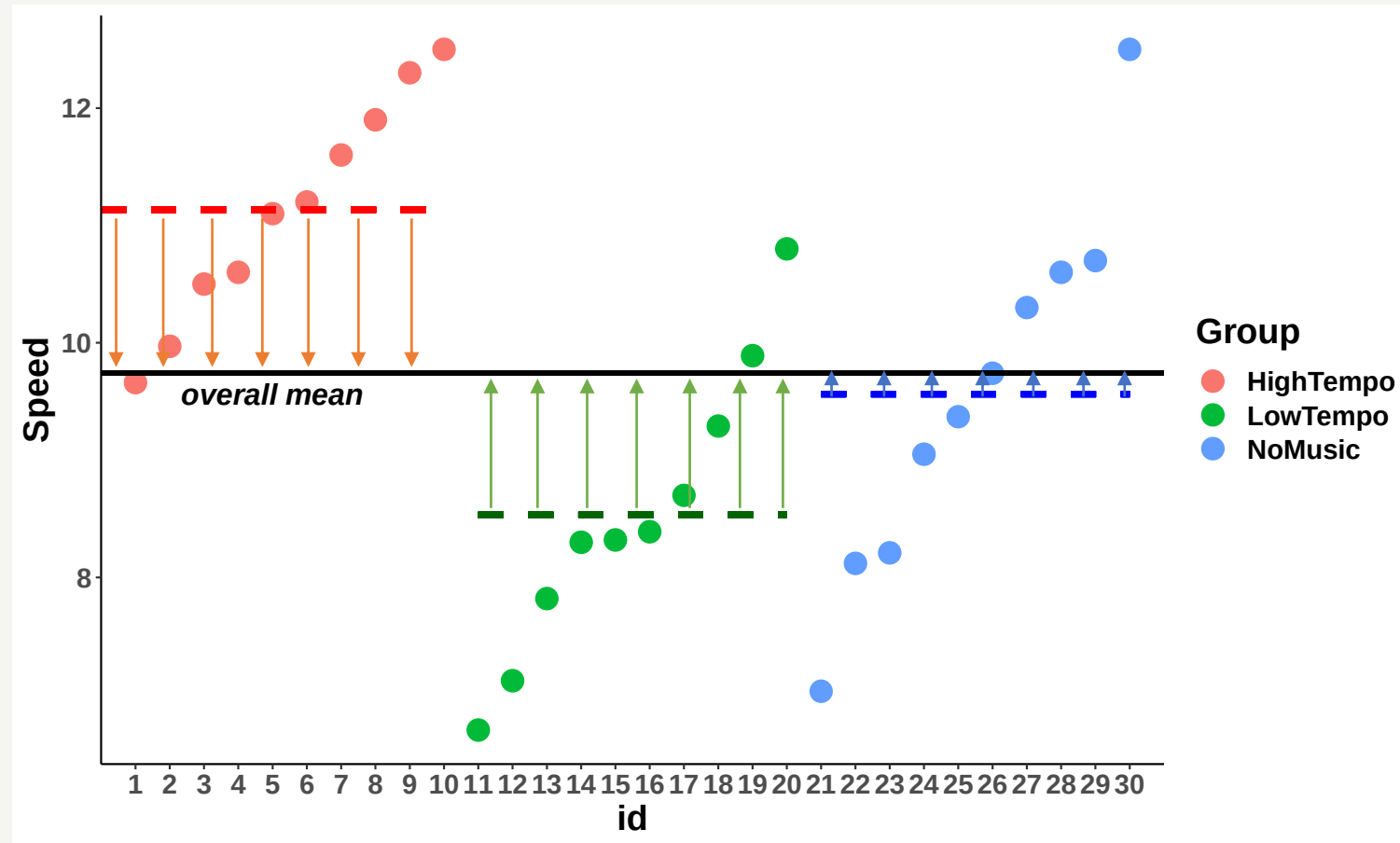
$$data = model + error$$

Sum of squared differences between
each data-point and **overall mean**

Sum of squared of difference between
each data-point and **group mean**

Sum of squared differences between
each group mean and **overall mean**

SS_{model}



SS_{model}

Group k	Speed X_{ik}	Group Mean $\bar{X}_k$	$[\bar{X}_k - \bar{X}]$	$[\bar{X}_k - \bar{X}]^2$
HighTempo	11.6	11.13	$11.13 - 9.74 = 1.39$	1.93
HighTempo	12.3	11.13	$11.13 - 9.74 = 1.39$	1.93
HighTempo	9.66	11.13	$11.13 - 9.74 = 1.39$	1.93
LowTempo	8.7	8.53	$8.53 - 9.74 = -1.21$	1.46
LowTempo	6.7	8.53	$8.53 - 9.74 = -1.21$	1.46
LowTempo	9.89	8.53	$8.53 - 9.74 = -1.21$	1.46
NoMusic	9.37	9.56	$9.56 - 9.74 = -0.18$	0.03
NoMusic	10.3	9.56	$9.56 - 9.74 = -0.18$	0.03
NoMusic	10.7	9.56	$9.56 - 9.74 = -0.18$	0.03
Overall Mean	9.74		SS _{model} =	34.29

Step 1: Partition variance

$$SS_{total} = SS_{model} + SS_{error}$$

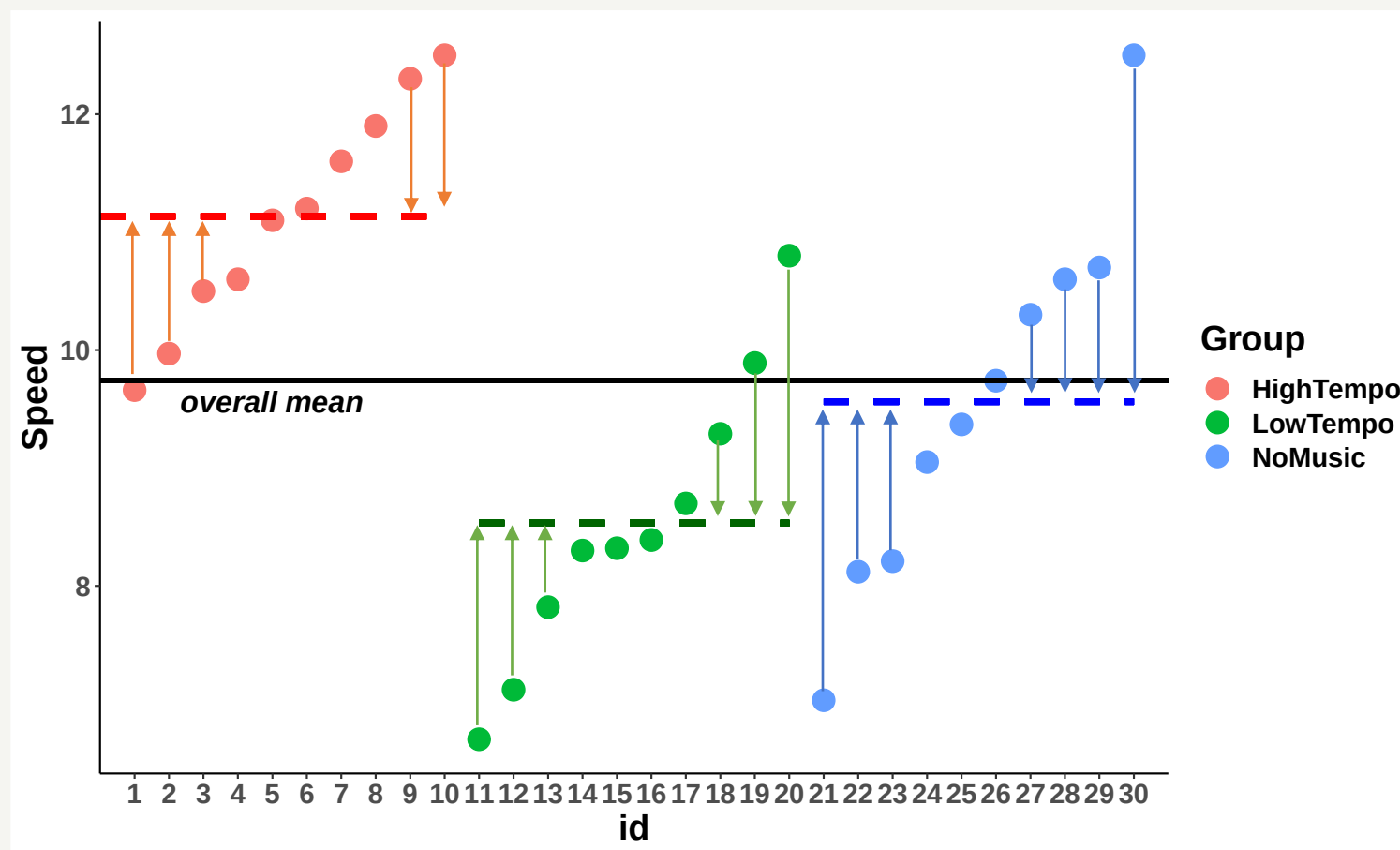
$$data = model + error$$

Sum of squared differences between
each data-point and **overall mean**

Sum of squared differences between
each group mean and **overall mean**

Sum of squared of difference between
each data-point and **group mean**

SS_{error}

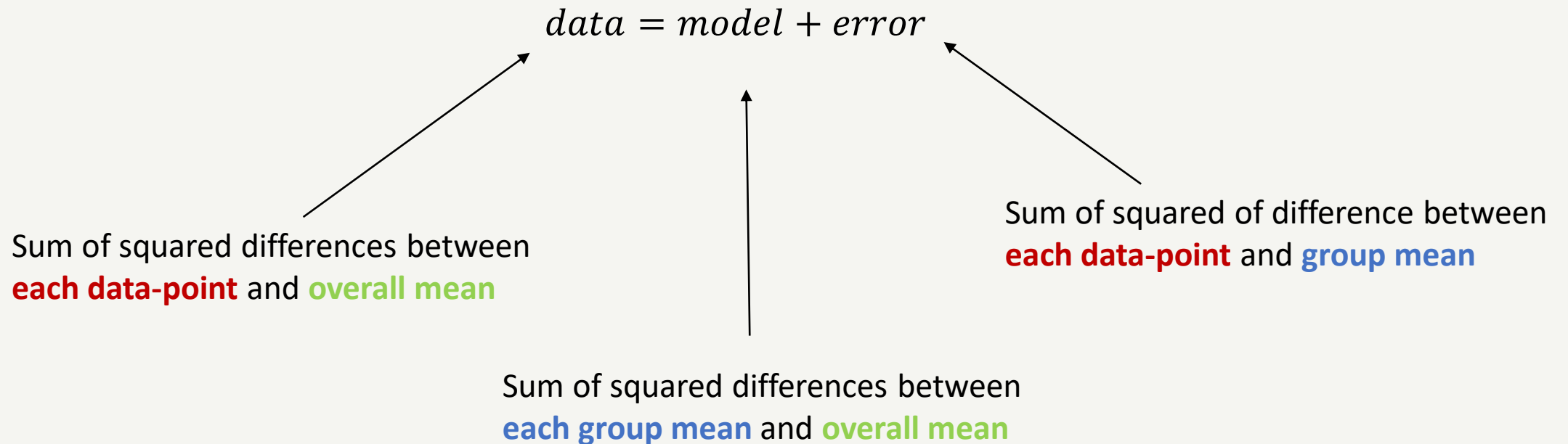


SS_{error}

Group k	Speed X_{ik}	Group Mean $\bar{X}_k$	$[\bar{X}_k - X_{ik}]$	$[\bar{X}_k - X_{ik}]^2$
HighTempo	11.6	11.13	11.13-11.6 = -0.47	0.22
HighTempo	12.3	11.13	11.13-12.3 = -1.17	1.37
HighTempo	9.66	11.13	11.13-9.66 = 1.47	2.16
LowTempo	8.7	8.53	8.53-8.7 = -0.17	0.03
LowTempo	6.7	8.53	8.53-6.7 = 1.83	3.35
LowTempo	9.89	8.53	8.53-9.89 = -1.36	1.85
NoMusic	9.37	9.56	9.56-9.37 = 0.19	0.04
NoMusic	10.3	9.56	9.56-10.3 = -0.74	0.55
NoMusic	10.7	9.56	9.56-10.7 = -1.14	1.30
Overall Mean	9.74		$SS_{\text{error}} =$	44.01

Step 1: Partition variance

$$SS_{total} = SS_{model} + SS_{error}$$
$$78.30 = 34.29 + 44.01$$



Does the model fit the data?

$$F = \frac{MS_{model}}{MS_{error}} = \frac{\left(\frac{SS_{model}}{DF_{model}} \right)}{\left(\frac{SS_{error}}{DF_{error}} \right)}$$

Now let's find the DFs of the model and the error

Step 2: Find DFs of model and error

Degrees of Freedom of the model (numerator) = number of groups - 1

$$3 - 1 = 2$$

Degrees of freedom of the error (denominator) = number of participants - number of groups

$$30 - 3 = 27$$

Step 3. Calculate F

$$F = \frac{MS_{model}}{MS_{error}} = \frac{\left(\frac{SS_{model}}{DF_{model}}\right)}{\left(\frac{SS_{error}}{DF_{error}}\right)} = \frac{\left(\frac{34.29}{2}\right)}{\left(\frac{44.01}{27}\right)} = \frac{17.15}{1.63} = 10.52$$

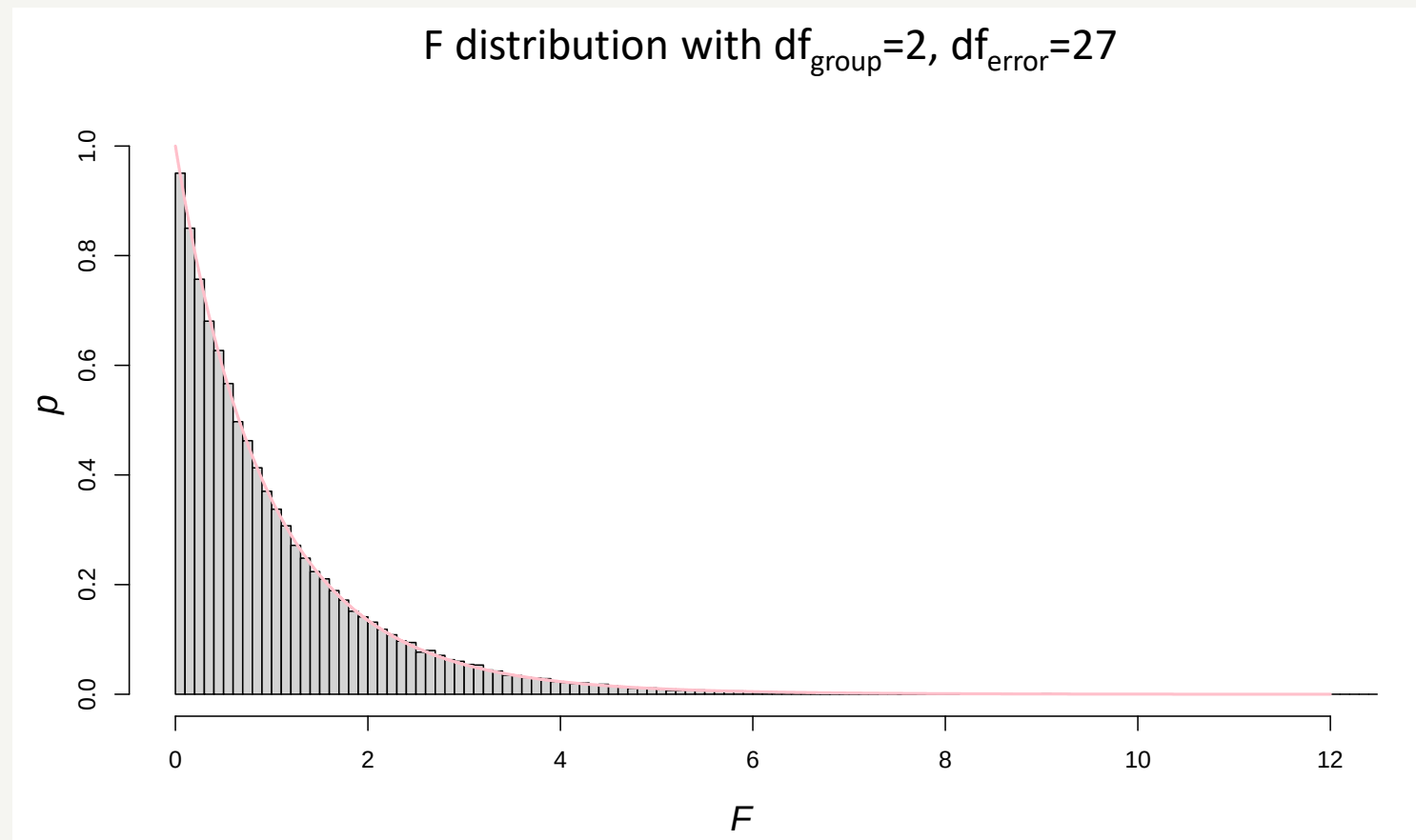
Reminder: Hypothesis testing

Steps:

1. Determine the null and alternative hypotheses
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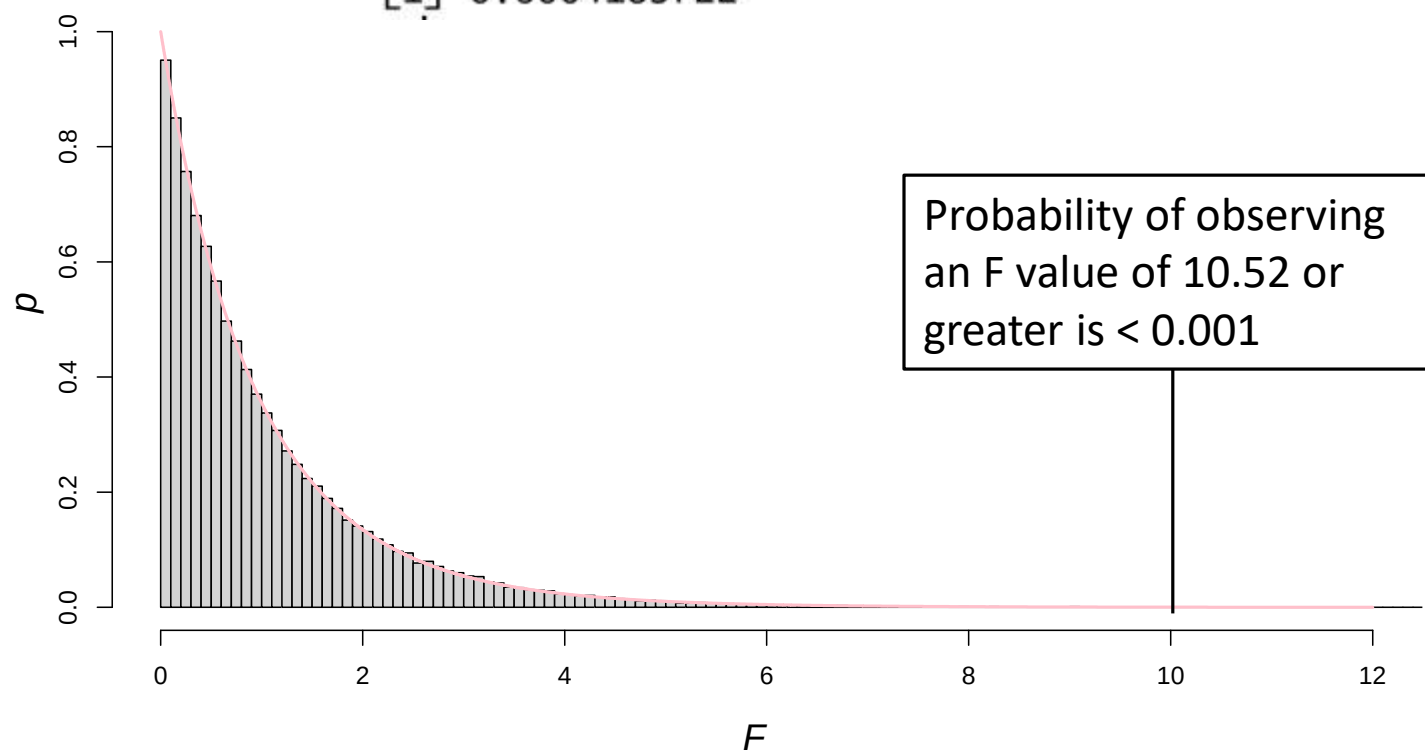
Null hypothesis: F distribution

- The distribution tells us the probability of observing F values under the null hypothesis (groups drawn from same population).
- F statistic is always positive as it is the ratio of two variances (which are always positive)
- The shape of the F distribution depends on the two degrees of freedom



Determining the p-value

```
> pf(10.52, 2, 27, lower.tail = FALSE)  
[1] 0.0004185722
```



One-Way ANOVA in R

One-Way ANOVA in R

Data must be in **long** format
(groups/conditions in one column,
scores in another)

Group	Speed	id
HighTempo	9.66	1
HighTempo	9.97	2
HighTempo	10.50	3
HighTempo	10.60	4
HighTempo	11.10	5
HighTempo	11.20	6
HighTempo	11.60	7
HighTempo	11.90	8
HighTempo	12.30	9
HighTempo	12.50	10
LowTempo	6.70	11
LowTempo	7.12	12
LowTempo	7.82	13
LowTempo	8.30	14
LowTempo	8.32	15
LowTempo	8.39	16
LowTempo	8.70	17
LowTempo	9.29	18
LowTempo	9.89	19
LowTempo	10.80	20
NoMusic	7.03	21
NoMusic	8.12	22

One-Way ANOVA in R

We are going to run an ANOVA using the package 'afex'

```

```{r}
library(afex)
model <- aov_ez(id = "id", # the column containing the subject IDs
 dv = "Speed", # the DV
 between = "Group", # the between-subject variable
 data = data)
anova(model)
```

```

Contrasts set to contr.sum for the following variables: Group

Anova Table (Type 3 tests)

Response: Speed

| | num | Df | den | Df | MSE | F | ges | Pr(>F) |
|-------|-----|----|-----|----|--------|--------|--------|---------------|
| Group | 2 | | 27 | | 1.6301 | 10.517 | 0.4379 | 0.0004192 *** |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Effect size

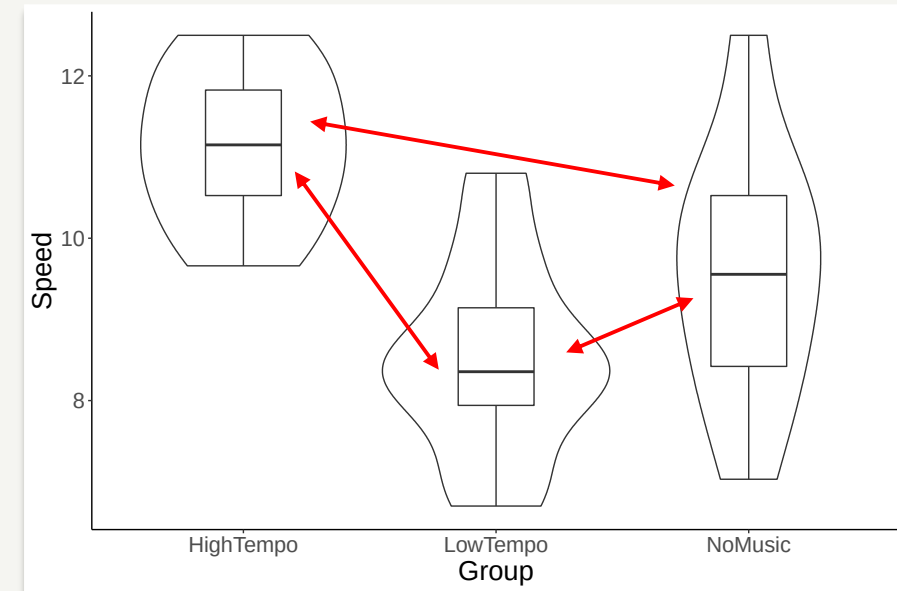
- **Eta squared** (η^2) is the effect size used with the F statistic
- Gives the proportion of total variation that is explained by our model (group membership) relative the total variation

$$\eta^2 = \frac{SS_{model}}{SS_{total}} = \frac{34.29}{78.30} = 0.44$$

- So, in our study 44% of variability is explained by music group condition

Post-hoc analysis

- **Important:** A significant F-test only tells us that differences between groups are most likely not due to chance
 - But it doesn't tell us *where* the difference between the groups lies: *which condition* is better for running speed?
- We need to perform follow-up tests to find out
- **Pairwise t-tests** to compare each group with each other
 - E.g. perform 3 t-tests to compare 3 groups: A:B, B:C, A:C
- But remember: each test comes with a 5% false-positive rate
 - We therefore need to **correct** the p-values for **multiple comparisons**



Bonferroni correction

- Multiply each p-value by number of comparisons.
 - E.g., p-value from a t-test is .025 but it's one of 10 tests → it would be corrected to 0.25 → would no longer pass our significance threshold
- Pairwise comparisons can be performed with Bonferroni correction in R using `pairwise.t.test()`

Comparing groups

```
pairwise.t.test(data %>% pull(Speed), data %>% pull(Group),
p.adjust.method = 'bonferroni')
```

```

Pairwise comparisons using t tests with pooled SD

data: data %>% pull(Speed) and data %>% pull(Group)

	HighTempo	LowTempo	
LowTempo	0.0003	-	} Bonferroni corrected p-values
NoMusic	0.0314	0.2481	

P value adjustment method: bonferroni

# Assumptions

Similar to the assumptions for independent sample t-tests:

1. The DV is interval or ratio data level
2. Independent observations
3. Homogeneity variance of residuals
  1. Can test using levene's test on residuals
4. Normal distribution of residuals
  1. Test using QQ-plot and shapiro-wilk test

ANOVA is robust to violations of these assumptions *if* group sizes are equal



# Reporting ANOVA results

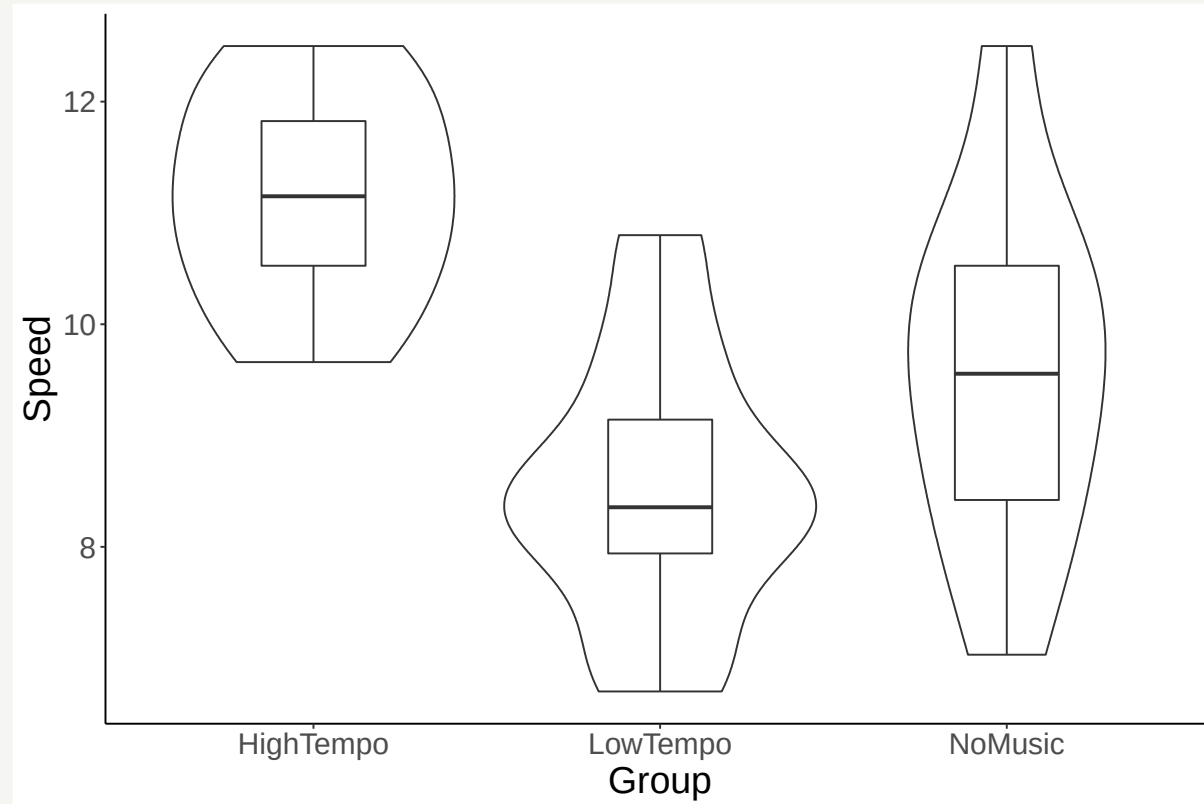
We conducted a one-way ANOVA on running speed (km/hr) with Group (three levels: High Tempo, Low Tempo, and No Music) as a between-subject variable. There was a significant main effect of group,  $F(2, 27) = 10.52$ ,  $p < .001$ ,  $\eta^2 = .44$ .

Post-hoc comparisons using Bonferroni correction showed that running speed was significantly higher in the High Tempo music group ( $M = 11.13$ ,  $SD = 0.96$ ) compared to the Low Tempo music group ( $M = 8.53$ ,  $SD = 1.23$ ,  $p < .001$ ) and the No Music group ( $M = 9.56$ ,  $SD = 1.57$ ,  $p = .031$ ). The Low Tempo and No Music group did not differ significantly in running performance ( $p = .248$ ).

# Write-up checklist

- Report type of ANOVA performed (“... one-way ANOVA ...”)
- Report the DV and IV (including levels) (“... speed (km/hr) with Group (three levels: High Tempo, Low Tempo, and No Music) as a between-subject variable ...”)
- Comment on significance (“... There was a significant main effect of group ...”)
- Report  $F(df) = Fvalue, p=pvalue$  (“... $F(2, 27) = 10.52, p < .001$  ...”)
- Report effect size (eta squared) (“ $\eta^2 = .44$ ”)
- Report post-hoc comparisons and adjusted p-values (“... Post-hoc comparisons using Bonferroni correction showed that ...”)
- Describe the direction of the pairwise comparisons (“... running speed was significantly higher in the High Tempo music group compared to the Low Tempo music group ...”)
- Report exact p-values, unless  $p < .001$
- Report all values to 2 decimal places (except for p values)

# Visualise the data



Thank you